

International Islamic University Chittagong
Department of Computer Science and Engineering
Program: B.Sc. in CSE: Semester: 5th
Mid-Term Examination, Spring-2025

Course Code: CSE 3523 Course Title: Microprocessor, Microcontroller and Embedded System
Time: 1 hour 30 minutes Full Marks: 30

Answer the following Three (3) questions. Each question carries 10 marks.

Questions

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| 1. | a) | Describe the physical address calculation in the 8086 microprocessor.
Include how segment and offset registers are used.
Find the physical address of 1000:2000. | 2 |
| | | | CO1 |
| 1. | b) | Write short notes on the following:
i) CISC
ii) RISC
iii) ALU | 3 |
| | | | CO1 |
| 1. | c) | Explain in detail the steps involved in the execution cycle of a microprocessor. | 5 |
| | | | CO2 |
| 2. | a) | Draw the internal block diagram of the 8086 microprocessor. | 4 |
| | | | CO1 |
| 2. | b) | What are the addressing modes used in the following Instructions? If the instructions are incorrect, then explain the reasons and make them correct | 2 |
| | | | CO1 |

i. MOV AL, BL
ii. MOV AL, BX
iii. MOV AH, [BX]
iv. MOV CL, 3A8H

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| 2. | c) | Write a short assembly program to add two 16-bit hexadecimal numbers 20F1H and 3C72H using 8-bit addition. Also, explain how the ADC instruction works in this context. | 4 |
| | | | CO2 |
| OR, | | | |
| 2. | c) | Find the value of C, A, P, Z, S, O FLAGS after execution of this instruction. | 4 |

MOV AL, 3FH MOV BL, 2AH ADD AL, BL	CO1
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| 3. a) | Write about the advantages and disadvantages of low-level language. | CO2 | 2 |
| 3. b) | Write an assembly language program to set the rightmost 3-bits, to clear the leftmost 2-bits, and to Invert the bit position 5, 7 & 8 th of register value AX=FF0EH. | CO2 | 4 |
| 3. c) | Given, AX=8877 H and BX=79ABH. Interchange the value of AX and BX using PUSH and POP Operation. Draw the Table of AX, BX and SP Value after every instruction. Take the initial address, SP= 2009H. | CO2 | 4 |

OR

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|-------|--|------------|---|
| 3. c) | What is the operation of the registers in the 8086 microprocessor during 8-bit and 16-bit multiplication? Write assembly language for the equation $y=6 \times 8 \times 10H$. | CO2 | 4 |
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